Acquired Long QT Syndrome (aLQTS): A Comprehensive Disease Characterization

MONDO: MONDO:1060292 · **Category:** Complex (non-Mendelian) · **Principal molecular target:** *KCNH2*/hERG (Kv11.1), the IKr potassium channel

Summary

Acquired Long QT Syndrome (aLQTS) is a complex, non-Mendelian disorder of ventricular repolarization in which a reversible external insult—most commonly a QT-prolonging drug, but also hypokalemia, bradycardia, hypomagnesemia, hypocalcemia, or metabolic/structural cardiac stress—prolongs the heart-rate-corrected QT interval (QTc) and predisposes to the polymorphic ventricular tachycardia **torsades de pointes (TdP)**, which can degenerate into ventricular fibrillation and sudden cardiac death. Unlike congenital LQTS, aLQTS has **no single causal gene**; the dominant molecular lesion is **pharmacological blockade of the rapidly activating delayed rectifier potassium current (IKr)**, carried by the channel encoded by **KCNH2 (hERG, Kv11.1; HGNC:6251)**. More than 100 structurally diverse drugs block hERG, and essentially every drug ever withdrawn for TdP liability is a hERG blocker (PMID: 16253929).

The central organizing principle of aLQTS is **"reduced repolarization reserve"** (Roden): normal repolarization is buffered by multiple redundant potassium currents, so subclinical lesions—latent congenital mutations, common modifier alleles, electrolyte disturbance, bradycardia, structural heart disease—do not prolong QT at baseline but unmask marked QT prolongation and TdP when an IKr-blocking drug is added (PMID: 16336514). This framework explains the idiosyncratic, patient-specific nature of the disease and frames aLQTS as a prototypical **gene–environment interaction**. Roughly **25–30% of aLQTS patients carry latent congenital LQTS mutations**, enriched in *KCNH2* (PMID: 26715165), and common variants such as **KCNE1 D85N** confer an odds ratio of ~9 for drug-induced LQTS (PMID: 22100668). Female sex and electrolyte abnormalities are the leading clinical risk factors.

Because the trigger is largely external and the ion-channel block is reversible, aLQTS is **highly preventable and treatable**: remove the culprit drug, correct electrolytes (keep potassium high-normal), give IV magnesium for TdP, and provide rate support (isoproterenol/pacing) for bradycardia-dependent episodes (PMID: 26183037). Validated risk scores (Tisdale), clinical decision support, and curated drug lists (CredibleMeds/AZCERT) support prevention in hospital settings (PMID: 36527915; PMID: 22549576).

Section 1 — Disease Information

Overview. Acquired Long QT Syndrome is a secondary (non-inherited) prolongation of ventricular repolarization, measured on the surface ECG as a prolonged QTc, that carries a risk of TdP. It is distinct from congenital LQTS in that it is precipitated by identifiable, usually reversible, external factors—predominantly medications. The core causal target is the **KCNH2/hERG (Kv11.1)** channel conducting IKr (PMID: 16253929).

Key identifiers. - **Mondo:** MONDO:1060292 - **Synonyms:** acquired LQTS (aLQTS); drug-induced long QT syndrome (diLQTS); secondary long QT syndrome; acquired QT prolongation - **MeSH:** Long QT Syndrome (D008133); related: Torsades de Pointes (D016171) - **ICD-10:** I45.81 (Long QT syndrome); drug-induced forms often coded with the offending agent (T-codes) plus I45.81 - **OMIM / Orphanet / causal gene:** Not applicable in the Mendelian sense — aLQTS has no single OMIM gene entry; the congenital counterparts (LQT1 *KCNQ1*, LQT2 *KCNH2*, LQT3 *SCN5A*) are the genetic reservoir from which susceptibility is drawn.

Data provenance. Evidence is drawn from both **aggregated disease-level resources** (systematic reviews, pharmacovigilance databases such as FDA FAERS, curated drug lists) and **individual-patient data** (EHR cohorts, case-control genetic studies, case reports). Both sources are heavily represented in the literature reviewed.

Section 2 — Etiology

Disease causal factors. The primary cause is **pharmacological**: drugs that block the hERG/IKr channel. Beyond direct pore block, some drugs act by **disrupting hERG trafficking** to the plasma membrane or accelerating its degradation (pentamidine, arsenic trioxide, probucol, geldanamycin) (PMID: 24998878). Non-pharmacologic acquired causes include electrolyte

derangements, bradyarrhythmias, hypothyroidism, hypothermia, starvation/anorexia, CNS injury (subarachnoid hemorrhage), Takotsubo cardiomyopathy, and myocardial ischemia/heart failure.

Genetic risk factors. aLQTS is polygenic/multifactorial on top of an environmental trigger: - **Latent congenital mutations** in *KCNH2*, *KCNQ1*, *SCN5A* (and minor genes) — present in ~28% of aLQTS patients (PMID: 26715165). - **Common susceptibility alleles**, most robustly **KCNE1 D85N (rs1805128)**, OR 9.0 (95% CI 3.5–22.9) (PMID: 22100668); additional candidate polymorphisms include NOS1AP variants and ethnic-specific alleles (e.g., *KCNH2* P448R/A915V in Asians; *SCN5A* S1102Y in African Americans) (PMID: 16540748).

Environmental/clinical risk factors. QTc >500 ms or ΔQTc ≥60 ms from baseline, advanced age, **female sex**, acute myocardial infarction, heart failure with reduced ejection fraction, hypokalemia, hypomagnesemia, hypocalcemia, bradycardia, diuretic use, elevated drug levels from drug–drug interactions (CYP inhibition), inadequate renal dose adjustment, and rapid IV administration (PMID: 27212965).

Protective factors. The inverse of the risk factors: maintaining potassium in the high-normal range, avoiding culprit drugs and QT-drug combinations, and androgenic hormonal milieu (dihydrotestosterone/androgens shorten repolarization, which may underlie lower male risk) (PMID: 27595633). No validated protective genetic variant is established.

Gene–environment interaction. aLQTS is the canonical cardiac GxE disorder: a subclinical genetic/electrolyte lesion reduces repolarization reserve, and only the combination with an IKr-blocking drug exceeds the threshold for clinical QT prolongation and TdP (PMID: 16336514).

Section 3 — Phenotypes

The core phenotype is **prolonged QTc on 12-lead ECG**; QTc >500 ms strongly supports pathology and ΔQTc ≥60 ms from baseline is significant (PMID: 41266879). Downstream phenotypes derive from the arrhythmia.

Phenotype	Type	HPO term	Onset / course	Frequency
Prolonged QT interval	Laboratory/ECG sign	HP:0001657	Acute, drug-triggered	Defining feature
Torsades de pointes	Clinical sign (polymorphic VT)	HP:0004755	Episodic	Rare but serious; up to ~3.6% for methadone
Syncope / presyncope	Symptom	HP:0001279	Episodic, arrhythmogenic	Variable
Seizure / seizure-like episodes	Symptom (can mimic epilepsy)	HP:0001250	Episodic	Can be misdiagnosed as epilepsy
Ventricular fibrillation / cardiac arrest	Clinical sign	HP:0001663	Acute	Rare
Sudden cardiac death	Outcome	HP:0001645	Acute	Rare
Arrhythmia / palpitations	Symptom/sign	HP:0011675	Episodic	Variable

Many patients are **asymptomatic until a trigger** is introduced. Seizure-like episodes are a notable mimic that can delay correct diagnosis (PMID: 23412014; PMID: 41266879). **Quality of life** is generally unaffected once the trigger is removed, as the condition is usually reversible; the burden is dominated by the acute arrhythmic risk rather than chronic disability.

Section 4 — Genetic / Molecular Information

Causal gene (molecular target). KCNH2 (hERG, Kv11.1; HGNC:6251; chromosome 7q36.1; NCBI Gene 3757) encodes the pore-forming subunit of the IKr channel — the dominant drug target in aLQTS (PMID: 16253929). Secondary contributing genes: *KCNQ1* (IKs), *SCN5A* (late INa), *KCNE1*/*KCNE2* (β-subunits).

Latent pathogenic variants. In an international cohort of 188 aLQTS probands screened across the five major LQTS genes, **53 (28%) carried 47 disease-causing mutations**. In "true aLQTS" (normal baseline QTc off triggers), **KCNH2 mutations were much more frequent than KCNQ1 (64% [95% CI 43–82%] vs 20% [95% CI 7–41%], P < 0.01)**. The control QTc of aLQTS patients (453±39 ms) was intermediate between congenital LQTS (478±46 ms) and non-carriers (406±26 ms) (PMID: 26715165). Variant classes are predominantly **missense** loss-of-function (K-channel) or gain-of-function (Na-channel), classified per ACMG/AMP.

Common modifier variant. KCNE1 D85N (rs1805128) predicted diLQTS with **OR 9.0 (95% CI 3.5–22.9)**; variant allele present in 8.6% of cases, 2.9% of drug-exposed controls, 1.8% of population controls, replicated in a validation cohort (3.5% cases vs 1.4% controls) (PMID: 22100668). KCNE1 encodes the MinK β-subunit modulating IKs.

Functional consequences. Potassium-channel variants → **loss of function** of repolarizing current; sodium-channel variants (*SCN5A*) → **gain of function** (sustained late inward sodium current). Trafficking-disrupting drugs reduce **membrane hERG expression** independent of pore block (PMID: 24998878).

Origin. Germline (inherited susceptibility variants); the disease itself is **not somatic**. **Epigenetic and chromosomal abnormalities** are not established contributors to aLQTS.

Section 5 — Environmental Information

Pharmacological (dominant environmental factor). See Section 12 culprit-drug table. Over 100 structurally diverse drugs block hERG; chemical-motif analysis of >300,000 compounds shows why structurally unrelated agents share this liability (PMID: 16253929; PMID: 25700001).

Electrolyte/metabolic. Hypokalemia (graded dose-response with severe QT prolongation, P-trend <0.001), hypomagnesemia, and hypocalcemia (PMID: 41397500; PMID: 35103080).

Lifestyle / physiologic. Bradycardia, starvation/anorexia nervosa, liquid protein diets, hypothermia. **Infectious agents:** not a direct cause, though infection-associated QT prolongation occurs via drugs, fever, and electrolyte shifts (e.g., COVID-19 treatment cohorts) (PMID: 33963634).

Section 6 — Mechanism / Pathophysiology

Ordered causal chain

```
1. External trigger (IKr/hERG-blocking drug, OR trafficking-disrupting drug,
   OR hypokalemia/hypomagnesemia/bradycardia)
        │  leads to
2. Reduction of net repolarizing K+ current (IKr), ± reduced membrane hERG
   (trafficking block) — INFERRED to summate with any latent genetic lesion
        │  results in  (branch: amplified where repolarization reserve is already reduced —
        │               latent KCNH2/KCNQ1/SCN5A variant, KCNE1 D85N, female sex, structural dis
3. Prolongation of ventricular action potential duration (APD) and QT interval
        │  permits
4. Reactivation of L-type Ca2+ current (ICaL) and/or enhanced late INa during phase 2/3
        │  generates
5. Early afterdepolarizations (EADs) in mid-myocardial (M) cells
        │  triggers
6. Triggered activity → focal premature ventricular excitation
        │  propagates into (requires)
7. A substrate of inhomogeneous/dispersed repolarization (transmural dispersion;
   M-cells vs epi/endocardium) with a concave repolarization-boundary geometry
        │  produces
8. Functional reentry → TORSADES DE POINTES (polymorphic VT)
        │  may degenerate to
9. Ventricular fibrillation → syncope / cardiac arrest / SUDDEN CARDIAC DEATH
```

Detail

Molecular/biochemical defect. The upstream lesion is a **cardiac ion-channel defect**: reduced IKr through hERG/Kv11.1. Drugs bind promiscuously within the hERG inner cavity; chemical-motif analysis of >300,000 compounds identified many scaffolds enriched for hERG inhibition, explaining why structurally unrelated agents share this liability (PMID: 25700001). A parallel branch is **trafficking disruption**, reducing surface channel density (PMID: 24998878).

Cellular process. Prolonged APD allows **ICaL reactivation** / **late-INa persistence**, generating EADs. EADs are the *trigger*; a *substrate* of dispersed repolarization is required for reentry. "TdP is initiated by triggered activity due to early afterdepolarization (EAD) and maintained by reentry within a substrate of inhomogeneous repolarization" (PMID: 38504714). Computational tissue models show reentry forms only when EAD clusters create a **concave repolarization-boundary geometry** (PMID: 37515880; PMID: 39837062).

Experimental recapitulation. Human engineered heart tissues reproduce the mechanism: hERG blockade (E-4031) prolongs APD, and combining hERG inhibition with **hypokalemia** + **hypomagnesemia** produces VT with marked beat-to-beat instability (PMID: 41727060).

Upstream vs downstream. Upstream = channel block / reduced reserve → APD prolongation (ionic/cellular). Downstream = EAD → triggered activity → reentry → TdP (tissue-level). **Cell types:** ventricular cardiomyocytes, especially **mid-myocardial M-cells** (CL:0000746, cardiac muscle cell; CL:0002129, regular ventricular cardiac myocyte). **GO terms:** GO:0086009 (membrane repolarization), GO:0098915 (membrane repolarization during ventricular cardiac muscle cell action potential), GO:0086091 (regulation of heart rate by cardiac conduction), GO:0005251 (delayed rectifier potassium channel activity). **CHEBI:** potassium(1+) (CHEBI:29103), magnesium(2+) (CHEBI:18420).

Section 7 — Anatomical Structures Affected

- **Organ:** Heart (UBERON:0000948), specifically the **cardiac ventricles** (UBERON:0002082) and ventricular myocardium (UBERON:0002349). Body system: **cardiovascular** (UBERON:0004535). Secondary: brain (hypoxic injury from arrhythmia/syncope).
- **Tissue/cell:** Ventricular **cardiac muscle** (striated); affected cells are **ventricular cardiomyocytes**, with **mid-myocardial M-cells** central to transmural dispersion of repolarization (CL:0000746, CL:0002129).
- **Subcellular:** Plasma membrane ion channels (GO:0005886), sarcolemmal **voltage-gated potassium channel complex** (GO:0008076); endoplasmic reticulum/Golgi relevant to the hERG trafficking branch (GO:0005783, GO:0005794).
- **Localization:** Bilateral/diffuse ventricular involvement; no lateralization.

Section 8 — Temporal Development

- **Onset:** Acute to subacute, tied to drug exposure or metabolic insult. Can occur at **any age**; older adults are over-represented (median age ~72 in a Swedish TdP cohort) (PMID: 41553550), but cases span neonates/infants to the elderly (PMID: 41896159).

- **Progression/course: Episodic and reversible.** QT prolongation develops over hours to days after drug initiation/dose increase; TdP is typically self-limited or terminated by intervention. The condition is **not chronic** once the trigger is removed.
- **Remission:** Treatment-induced remission is the norm — QTc normalizes after drug withdrawal and electrolyte correction.
- **Critical period:** The vulnerable window is during active drug exposure, especially with concurrent electrolyte disturbance, bradycardia, or multiple QT drugs — the key opportunity for monitoring and intervention.

Section 9 — Inheritance and Population

Epidemiology. Incidence of QTc >500 ms varies widely by drug and setting (0–16.2% for methadone; TdP prevalence up to 3.6%) (PMID: 42735272). In a prolonged-QTc inpatient cohort (n=2,735), **89 (3%) developed TdP** (PMID: 26589717). In a Swedish population cohort of 762 TdP patients (50% female, median age 72), 59% had been dispensed ≥1 risk drug and 73% had a risk drug and/or risk diagnosis (PMID: 41553550).

Inheritance. aLQTS is **multifactorial/complex**, not Mendelian. Susceptibility is **polygenic** (latent rare variants + common modifiers) interacting with environment. Penetrance of latent variants is **incomplete and trigger-dependent**.

Sex ratio. Strong **female predominance** of TdP risk (140/188 probands female; ~50–55% of cohorts female but higher per-exposure risk) (PMID: 26715165; PMID: 27595633). Notably, controlled crossover studies (dofetilide, quinidine, ranolazine, verapamil) found **no sex difference in concentration-adjusted QTc prolongation**, implying the excess female risk reflects lower baseline repolarization reserve/hormonal state rather than greater drug-induced QT lengthening per exposure (PMID: 26324176).

Population demographics. Older adults, patients with heart failure/MI, and those on polypharmacy are over-represented. Ethnic-specific modifier alleles exist (*KCNH2* P448R/A915V in Asians; *SCN5A* S1102Y in African Americans) (PMID: 16540748).

Section 10 — Diagnostics

Principal test: 12-lead ECG with QTc measurement (Bazett/Fridericia/Framingham). Diagnostic support: **QTc >500 ms on ≥2 ECGs** and/or **ΔQTc ≥60 ms from baseline** (PMID: 41266879; PMID: 38982033). Continuous telemetry for at-risk inpatients.

Laboratory: serum potassium, magnesium, calcium; renal/hepatic function; thyroid function; drug levels where applicable. Correct hypokalemia/hypomagnesemia.

Risk stratification tools.

Tool	Variables	Output	Evidence
Tisdale score (0–23)	Age >68, female, loop diuretic, K+ ≤3.5, admission QTc ≥450, acute MI, sepsis, HF, ≥1/≥2 QT drugs	Low ≤6 / Mod 7–10 / High ≥11	Mortality OR 4.80 (7–10) and 11.51 (≥11) vs <7 in 92,383 inpatients (PMID: 36527915)
RISQ-PATH	EHR-derived clinical factors	Risk probability	Outperformed by ML in head-to-head (PMID: 41139036)
Machine-learning (XGBoost/ DNN)	EHR + ECG traces	AUROC 0.86–0.87	Superior to Tisdale/RISQ-PATH (PMID: 41139036; PMID: 42951254)

Genetic testing. Not routine for isolated drug-induced events, but **targeted LQTS gene panel (KCNQ1, KCNH2, SCN5A, KCNE1, KCNE2)** is warranted when baseline QTc is prolonged off triggers, there is a family history, or TdP occurred with minimal provocation — ~28% yield of latent mutations (PMID: 26715165). KCNE1 D85N genotyping is a research/risk-modifier tool (PMID: 22100668).

Differential diagnosis. Congenital LQTS (persistent QTc off triggers; family history), electrolyte-only QT prolongation, epilepsy (seizure-like syncope mimics), structural VT, Takotsubo. Distinguish by reversibility after trigger removal.

Section 11 — Outcome / Prognosis

Prognosis is generally excellent once the trigger is removed — QTc normalizes and arrhythmic risk resolves. The acute danger is TdP progressing to VF and sudden death. Mortality and length-of-stay track the Tisdale score: vs QTc-RS <7, mortality **OR 4.80 (95% CI 4.42–5.21) for score 7–10** and **11.51 (95% CI 10.23–12.94) for ≥11**, with length of stay rising ~0.7 day per point (PMID: 36527915). Severe QTc prolongation is associated with ~3-fold higher ventricular arrhythmia risk and ~2-fold SCD risk among antipsychotic/antidepressant users (PMID: 41397500). Prognostic factors: magnitude of QTc, number of risk factors/QT drugs, electrolyte status, structural heart disease, and presence of latent genetic susceptibility.

Section 12 — Treatment

Acute management of TdP (PMID: 26183037): 1. **Discontinue the culprit QT-prolonging drug(s)**. 2. **IV magnesium sulfate** — first-line for TdP, even if serum Mg is normal. 3. **Correct electrolytes** — keep potassium in the high-normal range; replete magnesium/calcium. 4. **Electrical cardioversion/defibrillation** for sustained/hemodynamically unstable episodes. 5. **Rate acceleration** — isoproterenol or transvenous pacing for recurrent/bradycardia-dependent (pause-dependent) TdP. 6. **Avoid** repolarization-prolonging antiarrhythmics (Class IA/III).

Direct quote: *"Immediate treatment of TdP is by intravenous administration of magnesium sulphate, terminating prolonged episodes using electrical cardioversion. In refractory cases of recurrent TdP, the arrhythmia can be suppressed by increasing the underlying heart rate using isoproterenol (isoprenaline) or transvenous pacing"* (PMID: 26183037). NCIT intervention concepts: Magnesium Sulfate Therapy, Electrical Cardioversion, Cardiac Pacing, Isoproterenol.

Pharmacogenomics / drug–drug interactions. CYP3A4/CYP2D6/CYP1A2 inhibition raises culprit-drug plasma levels and TdP risk (e.g., ritonavir increasing hydroxychloroquine/lopinavir exposure) (PMID: 36437717). Genotype-aware prescribing and KCNE1/latent-variant status inform personalized risk.

Culprit drug classes (all hERG/IKr blockers unless noted): Class IA/III antiarrhythmics (quinidine, disopyramide, sotalol, dofetilide, ibutilide, amiodarone); macrolides/fluoroquinolones (erythromycin, clarithromycin, azithromycin, moxifloxacin, ciprofloxacin); azole antifungals; antimalarials (hydroxychloroquine, chloroquine, halofantrine);

antipsychotics (haloperidol, thioridazine, quetiapine, ziprasidone); antidepressants (citalopram/escitalopram, TCAs); antiemetics (ondansetron, domperidone); methadone; ranolazine; oncology agents (arsenic trioxide, anthracyclines, TKIs) (PMID: 26926294). **Trafficking inhibitors:** pentamidine, arsenic trioxide, probucol.

Drug	Association metric	Source
Ranolazine	TdP OR 53.6 (95% CI 5.4–524)	PMID: 26589717
Dofetilide	TdP OR ~25	PMID: 26589717
Furosemide (via hypokalemia)	39.3% of TdP cases	PMID: 26589717
Methadone	27% of TdP cases	PMID: 26589717
SSRIs	19.1% of TdP cases	PMID: 26589717
Amiodarone	18% of TdP cases	PMID: 26589717

Section 13 — Prevention

aLQTS is **largely preventable**. The AHA/ACCF scientific statement *"Prevention of Torsade de Pointes in Hospital Settings"* (Drew et al., 2010) codifies the approach: identify at-risk patients (proarrhythmic drugs, electrolyte disturbances, bradyarrhythmias) for QT-interval monitoring, avoid/limit concurrent QT-prolonging drugs, and correct hypokalemia/hypomagnesemia (PMID: 20142454; PMID: 22549576).

- **Primary prevention:** curated avoidance lists (**CredibleMeds/AZCERT** known/possible/conditional TdP risk), preclinical hERG screening (ICH S7B/E14, CiPA), electrolyte stewardship, dose adjustment in renal impairment, and avoiding QT-drug combinations (PMID: 20047143).
- **Secondary prevention:** baseline and on-treatment ECG monitoring, telemetry for high-risk patients, **clinical decision support alerts**, and risk scores (Tisdale/RISQ-PATH/ML) (PMID: 41139036; PMID: 41208201).
- **Tertiary prevention / counseling:** when a latent congenital variant is unmasked, **genetic counseling and cascade screening** of relatives.

Section 14 — Other Species / Natural Disease

aLQTS is fundamentally an **induced** phenotype rather than a naturally occurring heritable disease, so there is limited "natural disease" literature in other species. The hERG channel and IKr are **evolutionarily conserved** across mammals, and drug-induced QT prolongation is well documented in **dog** (canine, NCBI Taxon 9615) and **rabbit** (Oryctolagus cuniculus, Taxon 9986) models used for preclinical cardiac safety. Orthologous gene: *KCNH2* (human NCBI Gene 3757) with orthologs in mouse (*Kcnh2*), rat, dog, and zebrafish (*kcnh2/zERG*). Zoonotic/transmission considerations do not apply.

Section 15 — Model Organisms

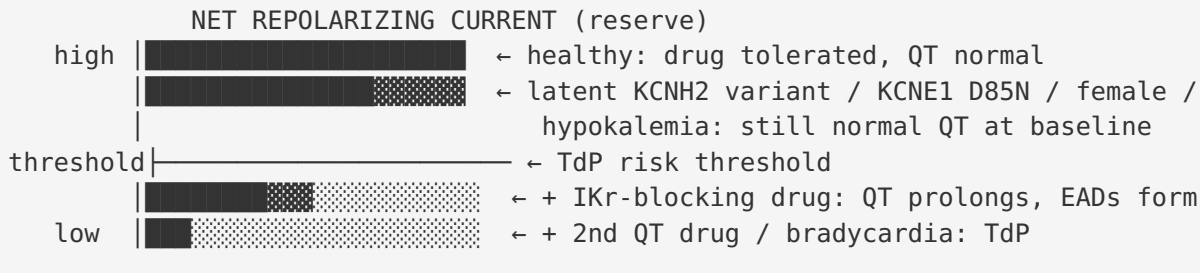
Model	Type	Use	Evidence
Heterologous hERG cells (HEK293/CHO)	In vitro	Patch-clamp hERG screening; ICH S7B/E14 standard	PMID: 20047143, PMID: 25700001
Human iPSC-CMs	In vitro / cellular	CiPA initiative; sex-specific repolarization-reserve studies	PMID: 30247688
Human engineered heart tissue (EHT)	3D in vitro	Recapitulates hERG-block + electrolyte-driven VT	PMID: 41727060
Zebrafish (Danio rerio)	Whole-organism	Acquired & inherited LQT2; delayed repolarization	PMID: 33519527
Rabbit/canine wedge, transgenic	Mammalian	Transmural dispersion, EAD/TdP	PMID: 38504714
In silico virtual populations	Computational	Sex-aware proarrhythmia prediction	PMID: 42216678

Phenotype recapitulation: sex-specific hiPSC-CMs reproduce the female repolarization-reserve deficit ("differential expression of KCNE1 gene and reduced repolarization reserve in female-derived hiPSC-CMs," PMID: 30247688); EHTs reproduce the EAD→VT cascade under combined hERG block + hypokalemia/hypomagnesemia; zebrafish reproduce whole-organ delayed repolarization ("Drug block of hERG channels or dysfunctional channel variants can

result in acquired or inherited LQTS2," PMID: 33519527). **Limitation:** no model fully captures the polygenic human susceptibility background, and small-animal repolarization currents differ from human.

Mechanistic Model / Interpretation

The entire disease collapses onto a single quantitative idea — **repolarization reserve** — pushed past a threshold:



Two independent "hits" are needed: a **substrate** (reduced reserve, often silent) and a **trigger** (drug/electrolyte). Upstream events are ionic (reduced IKr → longer APD); the pivotal cellular event is the **EAD**; the tissue-level requirement for lethality is **dispersed repolarization** enabling reentry. This explains why population-level QT prolongation is common but TdP is rare, why risk is idiosyncratic, and why prevention (removing the trigger) is so effective.

Evidence Base

PMID	Role in this report
16253929	Establishes KCNH2/hERG block as dominant mechanism; drugs withdrawn for TdP are hERG blockers
24998878	hERG trafficking-disruption branch of mechanism
26715165	28% latent congenital mutations; KCNH2 > KCNQ1 enrichment
22100668	KCNE1 D85N OR 9.0 — validated common susceptibility allele
16336514	Reduced repolarization reserve framework
27212965	Consolidated TdP risk-factor list
27595633 / 30247688	Female sex as independent risk factor; mechanistic basis
26324176	No sex difference in concentration-adjusted QTc — reserve, not drug-handling
36527915 / 38982033	Tisdale score and mortality gradient
38504714	EAD-trigger + reentry-substrate mechanism of TdP
41727060	Human EHT recapitulating hERG-block + electrolyte VT
26183037	Acute TdP management algorithm
26589717 / 26926294	Culprit-drug rankings and classes
20142454 / 22549576	AHA/ACCF prevention statement
25700001 / 33519527	hERG promiscuity; zebrafish model
16540748	Ethnic-specific modifier alleles; LQTS channelopathy review

Limitations and Knowledge Gaps

- **No single causal gene:** aLQTS is defined by an external trigger acting on a polygenic/environmental background; conventional OMIM-style gene annotation does not apply and the genetic architecture beyond KCNE1 D85N and latent LQTS variants is incompletely mapped.
- **TdP is rare and idiosyncratic**, making prospective quantification of absolute risk difficult; most data are from case-control, pharmacovigilance (reporting bias), and EHR cohorts rather than randomized designs.
- **Sex-difference mechanism** remains partly unresolved — crossover studies show no per-exposure QTc difference, pointing to baseline reserve/hormonal effects that are hard to quantify clinically.
- **Risk scores (Tisdale/RISQ-PATH)** have modest discrimination; ML models improve AUROC but need prospective validation and lack generalizability guarantees.
- **Trafficking-disruption pathway** is less well characterized than direct pore block and may be under-recognized in safety screening.
- **No validated protective genetic variant** is established.

Proposed Follow-up Experiments / Actions

1. **Prospective validation of ML risk models** (XGBoost + ECG-DNN) against Tisdale in multi-center inpatient cohorts, with calibration and net-benefit analysis.
2. **Polygenic risk score** combining KCNE1 D85N, NOS1AP, and latent rare LQTS variants to predict diLQTS, tested in drug-exposed EHR biobanks.
3. **Sex-aware iPSC-CM / EHT panels** to dissect the hormonal and KCNE1-expression basis of female susceptibility and to test whether androgen supplementation raises reserve.
4. **Systematic trafficking-liability screening** (surface-hERG assays) added to CiPA-style preclinical pipelines to capture non-pore-block mechanisms.
5. **Quality-improvement trials** of CDS alerts + electrolyte stewardship bundles to measure reduction in QTc >500 ms and TdP incidence.
6. **Cascade genetic screening protocols** for relatives of patients in whom a latent congenital variant is unmasked by aLQTS.

Report compiled from 14 confirmed findings and 51 reviewed papers across a 5-iteration autonomous investigation. Evidence types span human clinical cohorts, pharmacovigilance databases, in vitro/iPSC and engineered-tissue models, zebrafish, and computational simulation.